

IP Addressing in Detail

Lecture 4 | Organised Notes with Diagrams

This document reorganises the lecture notes on **IP Addressing** into a clean, point-wise format with explanations and diagrams for every concept: IPv4 & IPv6 structure, Network ID / Host ID, Address Classes (A–E), subnet mask basics, loopback address, broadcast address and real-world (private) IP examples.

1. What is an IP Address?

An **IP Address (Internet Protocol Address)** is a unique logical address assigned to every device connected to a network, so that it can be identified and can communicate with other devices. It works like a postal address for data packets on the internet.

There are two standard versions of IP addressing used today:

- **IPv4** – 32-bit address (older, still most widely used)
- **IPv6** – 128-bit address (newer, created to solve IPv4 address shortage)

2. IPv4 Addressing (32-bit)

IPv4 is a **32-bit** binary number, generally written in **dotted decimal notation** — the 32 bits are split into 4 equal parts of 8 bits each, called **octets**. Each octet is converted to a decimal number (0–255) and separated by dots.

32-bit IPv4 Address = 4 x 8-bit Octets



Example: 192 . 168 . 1 . 5

Fig 1: Structure of a 32-bit IPv4 address

Key range facts about IPv4:

- Minimum address: **0.0.0.0**
- Maximum address: **255.255.255.255**
- Total possible addresses = $2^{32} \approx 4.29$ billion
- **Loopback address:** 127.0.0.0 to 127.255.255.255 — reserved to let a device refer to itself (e.g. 127.0.0.1). Used for testing the local machine's own network stack.

3. IPv6 Addressing (128-bit)

IPv6 was introduced because IPv4's ~4.3 billion addresses were running out. It is a **128-bit** address, written in **hexadecimal** form instead of decimal.

- The 128 bits are divided into **8 groups of 16 bits** each ($8 \times 16 = 128$ bits).
- Each 16-bit group is written as a 4-digit hexadecimal number, and groups are separated by a colon (:).
- Hexadecimal digits used: 0–9 and A–F.

IPv6: 8 groups x 16 bits = 128 bits

0000	0000	0000	0000	0000	0000	0000	0000
------	------	------	------	------	------	------	------

Fig 2: 128-bit IPv6 address = 8 hextets of 16 bits

Example full-form IPv6 address range:

Minimum: **0000:0000:0000:0000:0000:0000:0000:0000**

Maximum: **FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF**

Total possible IPv6 addresses = 2^{128} , an enormously larger address space than IPv4.

4. Network ID and Host ID

Every IPv4 address is logically divided into two parts:

- **Network ID (NID)** — identifies which network the device belongs to.
- **Host ID (HID)** — identifies a particular device (host) within that network.

How many bits go to Network ID and how many to Host ID depends on the **Class** of the address (explained in Section 5) or on the **subnet mask** being used.

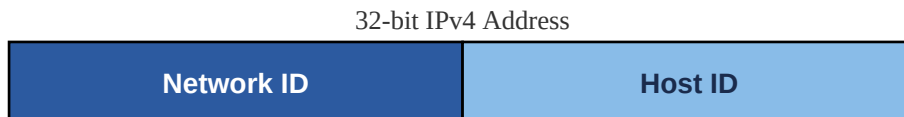


Fig 3: Every IP address splits into Network ID + Host ID

Broadcast Address: Within a network, the address in which all Host ID bits are set to 1 is reserved as the **broadcast address** — used to send data to every host on that network at once (it cannot be assigned to a single device).

5. Classful IP Addressing (Class A – E)

The original IPv4 addressing scheme divides the address space into 5 classes — A, B, C, D and E — based on the value of the **first octet** (and hence the leading bits of the address). Classes A, B and C are used for normal host addressing (with decreasing number of networks but increasing hosts per network); Class D is for multicasting; Class E is reserved for future / experimental use.

Class	1st Octet Range	Leading Bits	Default Mask	Network Bits	Host Bits	Purpose
A	1 – 126	0	255.0.0.0	8	24	Very large networks
B	128 – 191	10	255.255.0.0	16	16	Medium-sized networks
C	192 – 223	110	255.255.255.0	24	8	Small networks
D	224 – 239	1110	—	—	—	Multicasting
E	240 – 255	1111	—	—	—	Research / Future use

Fig 4: Summary table of IP address classes

Note: **127.x.x.x** (which numerically falls inside the Class A range) is **not** used for normal hosts — it is fully reserved for the Loopback address.

5.1 How the class ranges are derived from binary

Each class is recognised by the fixed leading bit pattern of the first octet. This is why the decimal ranges in the table above come out the way they do:

Class	Bit Pattern	Lowest Value	Highest Value	Rule
Class A	0 xxxxxxx	0 0000000 = 0	0 1111110 = 126	First bit fixed = 0
Class B	10 xxxxxx	10 000000 = 128	10 111111 = 191	First 2 bits fixed = 10

Class C	110 xxxxx	110 00000 = 192	110 11111 = 223	First 3 bits fixed = 110
Class D	1110 xxxx	1110 0000 = 224	1110 1111 = 239	First 4 bits fixed = 1110
Class E	1111 xxxx	1111 0000 = 240	1111 1111 = 255	First 4 bits fixed = 1111

(Note: 127 is skipped from Class A's usable range because it is reserved for loopback, so Class A's usable first-octet range is quoted as 1–126.)

5.2 Class-wise Network ID / Host ID split (diagram)

Since the total address is always 32 bits, giving more bits to the Network ID (as in Class A) means fewer networks but far more hosts per network — and vice-versa for Class C.

Class A -> 8 bits Network | 24 bits Host (few networks, huge hosts each)



Class B -> 16 bits Network | 16 bits Host (balanced)



Class C -> 24 bits Network | 8 bits Host (many networks, few hosts each)

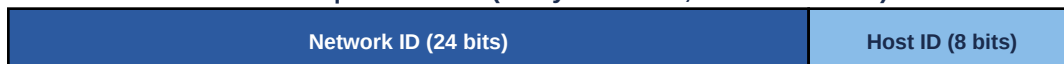


Fig 5: Network ID / Host ID proportion for Classes A, B and C

5.3 Number of networks and hosts per class

Class	Networks Available	Hosts per Network
A	$2^7 = 128$ (126 usable)	$2^{24} - 2 = 16,777,214$
B	$2^{14} = 16,384$	$2^{16} - 2 = 65,534$
C	$2^{21} = 2,097,152$	$2^8 - 2 = 254$

We subtract 2 from the host count because the all-0 host bits address is the **Network Address** itself, and the all-1 host bits address is the **Broadcast Address** — neither can be assigned to an actual device.

6. Real-World Example: Reading an IP Address

Take the example address **73.0.0.0** used by an ISP (Internet Service Provider, e.g. Airtel / Idea / Vodafone type connections in the notes):

Address	73 . 0 . 0 . 0
First octet	73 → falls in 1–126 range
Class	Class A
Network ID	73 (8 bits)
Host ID range	0.0.0 – 255.255.255 (24 bits)
Broadcast address	73.255.255.255

Similarly, an address such as **182.255.255.255** has first octet 182, which lies in the 128–191 range, so it belongs to **Class B**.

7. Quick Recap

- ✓ IPv4 = 32-bit address (4 octets), written in decimal, dot-separated.
- ✓ IPv6 = 128-bit address (8 hextets), written in hexadecimal, colon-separated.

- ✓ Every address = Network ID + Host ID.
- ✓ Classes A/B/C = normal host addressing; Class D = multicast; Class E = reserved/experimental.
- ✓ Class A: 1–126, Class B: 128–191, Class C: 192–223, Class D: 224–239, Class E: 240–255.
- ✓ 127.x.x.x is reserved for Loopback testing, not a usable Class A network.
- ✓ Broadcast address = network address with all Host ID bits set to 1.